SITE ANALYSIS REPORT

S3 | Assosa Preprimary | 134 Students | Option A

Total Students	134
Design Option	Option A (Full)
Existing Buildings	4 structures (640 m2 total)
Existing Trees	3 identified from aerial imagery
Adjacent Roads	13 road segments in KML data
Highway Proximity	Yes - B40 Highway frontage

1. Access

School S3 is located within the Assosa urban road network. The site is bounded by public roads that define its access character, ranging from 10m residential streets to major arterials such as the B40 Highway where applicable. Pedestrian access for children aged 5-6 years is the primary design concern: safe crossings, traffic calming near the school gate, and separation of vehicular and pedestrian flows are mandatory requirements under IRC safeguarding protocols. The main entry point should be clearly defined with a gated access that is child-proof yet adult-accessible, positioned away from the highest-traffic road frontage to minimize child exposure to moving vehicles. Emergency vehicle access must also be maintained throughout the construction and operational phases.

Critical access constraint: The B40 Highway frontage requires a minimum 3m vegetated buffer zone between the roadway and the playground fence to mitigate dust, noise, and vehicle proximity hazards. The buffer should use preserved or newly planted trees to create both a physical and visual barrier. School gates must not open directly onto the highway; a secondary internal pathway must lead children from the gate to the play area.

2. Circulation

Internal circulation at School S3 currently consists of informal movement patterns across bare earth or compacted dirt surfaces between buildings and open areas. The design must formalize these patterns into defined 1.2m-wide compacted gravel pathways (10-20mm gravel, 100mm depth) that connect the classroom buildings to the play area, the entry gate, and between different play zones. All pathways must be obstacle-free and wheelchair-accessible where terrain permits, complying with IRC inclusion standards.

The pathway network should employ a looping or circular configuration rather than dead-end routes, enabling continuous movement patterns and improving teacher supervision sight lines. A primary spine connects the building entrance to the active play zone, with a secondary branch leading to the quiet/shade area. All pathways must incorporate a minimum 1% drainage cross-fall to prevent ponding during the April-October wet season.

3. Topography

School S3 is situated on predominantly flat terrain, which is advantageous for construction with minimal grading costs. Equipment foundations can be excavated to uniform depths without requiring stepped or variable-height footings. However, the flat topography creates a critical drainage challenge: without engineered fall, rainwater will pond on play surfaces during the wet season (approximately 1,200mm annual rainfall concentrated in April-October).

The design must incorporate a minimum 1% artificial slope across all play surfaces, directing surface water toward perimeter swale drains (300mm x 300mm, rubble-filled, gravity-fed). The lateritic red soil (CBR 5-10%) has moderate permeability but will become impassable mud when saturated. Washed river sand surfacing under equipment

(150-300mm depth) will aid drainage while providing impact attenuation. All cut/fill operations, though minimal, must achieve 95% modified Proctor compaction.

4. Vegetation

School S3 has sparse existing vegetation with 3 or fewer trees on the entire site. The VLM analysis notes zero existing shade, meaning 100% of shade requirements must be met through new planting and, where necessary, temporary shade structures. This represents the highest planting priority among the six schools and requires an accelerated tree establishment programme.

Planting strategy: A minimum of 2 drought-tolerant shade trees (species such as Markhamia lutea, Cordia africana, or Ficus sycomorus) must be planted immediately at the start of construction to begin canopy establishment. Fast-growing species should be prioritized for the first 3-5 years, with longer-lived structural species interplanted for long-term canopy coverage. In the interim, shade can be provided by simple timber pole and thatch shelters that can be community-constructed. All new planting must use boron-based termite treatment (24-hour soak) for timber components.

5. Constraints

The design for School S3 is subject to multiple overlapping constraints that define the buildable envelope. Fixed constraints include all existing buildings (3.0m minimum clearance for swing/slide platforms, 1.5m minimum for general equipment), preserved trees (1.5m trunk protection radius), and the site boundary fence (1.5m minimum equipment setback). These non-negotiable setbacks collectively reduce the usable play area and must be accurately documented before any equipment placement.

Highway buffer: The B40 Highway frontage requires a 3m safety buffer in addition to the standard 1.5m fence setback, effectively creating a 4.5m exclusion zone along that boundary. This zone must be vegetated with preserved or new trees and cannot contain any play equipment or furniture. Noise and dust from highway traffic are significant environmental constraints that require the tree buffer as primary mitigation.

Material constraints: All materials must be sourced within 50km of Assosa. Banned materials include synthetic turf, rubber tiles (except recycled tire mulch), imported specialty items, and exotic invasive plant species. All foundations must be achievable by manual labour with basic tools (no heavy machinery), with maximum 0.05 m3 concrete per footing (hand-mixed C20). Installation crew capacity is 4 unskilled labourers plus 1 skilled metalworker per equipment item.

6. Noise

School S3 is exposed to significant noise from the adjacent B40 Highway, which carries moderate to heavy traffic throughout the day. Traffic noise is a particular concern during outdoor play periods, as prolonged exposure to elevated noise levels (above 55 dBA) can affect children's concentration, stress levels, and speech development. The VLM analysis explicitly flags this as requiring a dedicated noise and dust buffer zone.

Noise mitigation strategy: The primary noise buffer is the preserved tree belt along the highway frontage. Mature trees with dense canopy cover can reduce noise levels by 5-10 dBA through absorption and scattering. To enhance this effect, the buffer zone should be supplemented with additional planting (minimum 2 rows of trees at 3m spacing) using fast-growing, dense-canopy species. Play equipment placement should prioritize the quieter interior of the site, with quieter activities (reading, sand play) placed nearer the highway and noisier active play (shouting games, ball play) placed toward the interior.

7. Sun Path

Assosa is located at approximately 10 degrees North latitude, placing it in the tropical zone where the sun path is characterized by relatively small seasonal variation in solar altitude. At this latitude, the sun passes nearly overhead during the equinox periods (March and September), with a solar altitude of approximately 80 degrees at solar noon. During the June solstice, the sun reaches its zenith directly overhead (altitude approximately 90 degrees), while the December solstice sees a noon altitude of approximately 57 degrees.

Implications for School S3: The near-overhead midday sun means that vertical structures (buildings, trees) cast very small shadows at midday, providing minimal shade at the time of highest UV exposure. Shade planning must therefore rely on horizontal canopy cover (tree canopies, shade structures) rather than building shadows. The morning sun (east) and afternoon sun (west) strike at lower angles, creating longer shadows from vertical elements. East-side trees provide valuable morning shade during arrival time, while west-side planting is critical for afternoon play sessions when temperatures peak. Equipment should be oriented to avoid children facing directly into low-angle sun during peak play hours (typically 10:00-15:00).

8. Wind

Assosa experiences predominantly easterly and southeasterly winds during the wet season (April-October), driven by the Intertropical Convergence Zone (ITCZ) migration. During the dry season (November-March), winds shift to a northeasterly pattern influenced by the Harmattan. Wind speeds are generally light to moderate (5-15 km/h), though gusts can be higher during the transitional months of March-April and October-November.

Design implications for School S3: Wind direction analysis informs the placement of dust-sensitive play areas and the design of windbreaks where needed. Sand play areas and loose-surfaced zones should be positioned downwind of the prevailing dry-season direction (northeast) to minimize dust blown onto children. The preserved and proposed tree plantings serve a dual function as windbreaks, reducing wind speed across the play area by 30-50% depending on planting density and maturity. Equipment with large sail areas (merry-go-rounds, swings) should be oriented perpendicular to prevailing winds to minimize unwanted spinning or swinging from wind alone.

During the wet season, the easterly winds can drive rain at an angle, meaning that east-facing shelter (building eaves, tree canopies, shade structures) provides the most effective rain protection. The design should ensure that the quiet/shade zone and any seating areas have east-facing wind/rain protection for year-round usability.

9. Existing Land Use

School S3 is part of the Assosa City educational infrastructure, situated within an established urban neighbourhood. The surrounding land use is predominantly residential (R-1 and R-2 zoning as identified in the VLM analysis), with some mixed-use areas including institutional buildings (health centres, mosques, NGO offices such as IRC and Plan International as noted in the VLM data). The site itself is currently used for school operations with existing classroom buildings, open play areas, and peripheral circulation.

The existing play area consists of bare earth or overgrown grass with no formal play infrastructure. The ground surface is lateritic red soil typical of the Benishangul-Gumuz region, which becomes dusty in the dry season and muddy in the wet season. Converting this to safe play surfacing (washed river sand, compacted gravel, drought-tolerant grass) is a primary objective of the design. The surrounding residential context means that the playground will be visible from neighbouring properties, and play noise may be audible. The design should include perimeter planting to provide visual privacy and acoustic buffering where the playground abuts residential boundaries.

The proximity to the B40 Highway (a major arterial road) and the adjacent Cesca Health Center and Hidri Mosque (as identified in the VLM analysis) means the school sits at an intersection of institutional and transportation land uses. This creates both challenges (noise, dust, traffic safety) and opportunities (access to utilities, community visibility, potential shared-use arrangements with neighbouring institutions).